Bethany Burt

Doctoral Candidate in Astrophysics

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Research Interests

Rocky-planet atmosphere modelling • Radiative transfer & climate–chemistry feedback • Photochemistry • Star-planet-life connection • Atmospheric survival & habitability • Biosignature identification • Life in alternative solvents • Non-LTE atomic physics & stellar spectroscopy

Education

PhD in Astrophysics, expected 2029 October 2025–present

Ludwig Maximilian University of Munich, Munich, Germany

International Max Planck Research School on Astrophysics; University Observatory Munich and Munich Center for Geoastronomy

Advisor: Prof. Kevin Heng. Dissertation: "Islands of Stability: The Cosmic Archipelago"

MSci Chemistry with First-Class Honours September 2019–June 2024

University of Glasgow, Glasgow, United Kingdom

Research Positions

Doctoral Candidate October 2025–present

University Observatory Munich and Munich Center for Geoastronomy *Munich, Germany*

Advisor: Prof. Kevin Heng. Modelling rocky-planet atmosphere stability; collaborating with Prof. Sara Seager's Toronto group on biomolecular stability in alternative solvents.

Research Assistant September 2024–October 2025

Max Planck Institute for Astrophysics *Garching, Germany*

Supervisor: Prof. Selma de Mink. Analysis of carbon and nitrogen abundances in stripped stars using radiative-transfer code CMFGEN.

ISTernship Research Intern June–August 2025

Institute of Science and Technology Austria *Vienna, Austria*

Supervisor: Prof. Ylva Götberg. Continued the stripped-star project during an in-person research stay.

MSci Researcher September 2023–May 2024

University of Glasgow *Glasgow, United Kingdom*

Supervisors: Prof. Klaas Wynne and Dr. Mario González Jiménez. Ultrafast spectroscopy of pre-nucleation clustering and non-classical crystallisation.

Erasmus Research Student September 2022–May 2023

Max Planck Institute for Astronomy *Heidelberg, Germany*

Supervisor: Dr. Maria Bergemann. Created a molecular model of cyanogen for non-LTE radiative-transfer analysis of the solar nitrogen abundance.

Publications and Manuscripts

Under review

Burt, B., Heng, K., Cridland, A., & Tsai, S.-M. "Uncertainty and Disequilibrium Chemistry in C/O and Mean Molecular Weight." Under review at *Astronomy & Astrophysics*. arXiv: https://arxiv.org/abs/2610.24583

In preparation

Burt, B., Nicholls, H., Tsai, S.-M., & Heng, K. "The Fate of Sulfur in Warm Rocky Exoplanet Atmospheres: Coupled Climate–Chemistry and Observable Signatures."

Burt, B. & Seager, S. "Dipeptide Decomposition in Concentrated Sulfuric Acid: Reaction Mechanisms and Implications for Biomolecular Stability in Alternative Solvents."

Burt, B., Götberg, Y., Dutta, D., Laroche, A., Drout, M., & de Mink, S. E. "Into the Hearts of Stripped Stars: Carbon and Nitrogen Abundances as Tracers of CNO Processing."

Presentations

Conference and Workshop Talks

Beyond Water: Lipid Self-Assembly in Ionic Liquids. Astrobiology Graduate Conference (AbGradCon), University of Arizona, Tucson, Arizona. Selected oral presentation. — September 2026

Continuum Fluorescence and the Carbon Abundance Problem in Stripped Stars. Fourth X-Shooting ULLYSES Workshop, Institute of Science and Technology Austria, Klosterneuburg, Austria. — March 2025

Project and Departmental Talks

Searching for Stable Atmospheres. Geoastronomy Synergy ERC Strategy Meeting, Tenerife, Spain. — September 2026

Atmospheric Diagnostics: Which Is Less Wrong? Geoastronomy Synergy ERC Strategy Meeting, Herrsching, Germany. — March 2026

Light Matters: Photochemistry and Radiative Transfer. Geoastronomy Synergy ERC Strategy Meeting, Helsinki, Finland. — September 2025

Investigating Non-Classical Nucleation with Optical Kerr Effect Spectroscopy. Physical Chemistry Seminar, School of Chemistry, University of Glasgow. — March 2024

Posters

The Life We Haven't Looked For (Yet). Exoplanets 6, Porto, Portugal. — June 29–July 3, 2026

Beyond Water: Lipid Self-Assembly in Ionic Liquids. Astrobiology Science Conference (AbSciCon), Madison, Wisconsin. — May 2026

Atmospheric Diagnostics: Which Is Less Wrong? Effects of Disequilibrium Chemistry on Mean Molecular Weight and C/O. Ariel: Science, Mission & Community 2026, Harwell, United Kingdom. — March 2026

What Physics Is Missing from Exoplanet Atmosphere Models? ORIGINS Science Week, Kloster Seeon, Germany. — December 2025

Into the Hearts of Stars: Carbon and Nitrogen Abundances of Stripped Stars. ISTern Fellow Poster Session, Institute of Science and Technology Austria, Klosterneuburg, Austria. — August 2025

Building a Non-LTE Molecular Model of CN for Solar Spectroscopy. MPIA Advisory Board Visit, Max Planck Institute for Astronomy, Heidelberg, Germany. — February 2023

Teaching, Leadership, and Service

Teaching and Mentoring

Undergraduate Mentorship: Amina A. (2023–2026), Katarina C. (2024–2025), Thomas D. (2025–2026), Natalie W. (2025–2026), Shawn A. (2024–2026), Angeline G. (2025–2026) — 2023–present

Private tutoring (700+ lessons): 1 adult learner, 40 high-school students — 2021–present

High-School Exam Support Volunteer, University of Glasgow — October 2023–May 2024

Leadership and Service

Local Organizing Committee, 21st IMPRS Symposium, Garching, Germany — May 7, 2026

Head Activity Counsellor, Camp Stewart for Boys, Texas — Summers 2022 and 2023

Professional Development

Workshops and Courses

Practical Research Computing Toolkit Part 1: Bash, Python Environments, and Git for Early-Career Scientists. IMPRS, Garching, Germany. — January 12–16, 2026

Astronomical Data: Overview and Practical Examples IMPRS, Garching, Germany. — January 26–30, 2026

Summer Schools

Exoplanets by the Lake IV: Geoastronomy and Origins of Life, Herrsching am Ammersee, Germany. — July 2026

Exoplanets by the Lake III: Atmospheres and Beyond, Herrsching am Ammersee, Germany. — July–August 2025

Awards and Research Support

AbGradCon 2026 travel support ($750) — September 2026

Travel support for MIT research visit (approximately $4,500), financed by the Seager lab. — July 2026

ISTA Austria: ISTernship Summer Programme Scholarship (Non-EU Countries) (€3,385), awarded by OeAD and financed by ISTA Austria. — June–August 2025

The School of Chemistry Organic Level 3 Prize 2022 (£200), University of Glasgow. — May 2023

Erasmus+ funding, MPIA research placement (€6,210). — September 2022–May 2023

Community Science

Endorser and workshop participant for Kiang, N. Y., et al. (2026), "White Paper on Phototrophic Biosignatures: Research Priorities for the Search for Life on Other Worlds." arXiv:2607.03106. — 2026